Behavioral Phenotyping of Sleep and Daily Activity Across Age and Gender Using Free-Living Wrist-Worn Accelerometer Data

Md Sahriar Asif, Lubaba Hasan, Tasnim Abrar Sajin

Department of Computer Science and Engineering, University of Asia Pacific (UAP), Dhaka, Bangladesh

1 Introduction

Sleep and physical activity are established determinants of cardiovascular and metabolic health, yet guidelines assume everyone in an age group behaves alike. Wrist-worn accelerometers now measure real-life behavior objectively, but research has mostly classified single activities or total time in intensity levels.

147	19	3
participants analysed	behavioral features	phenotypes found

2 Problem Statement

- Do temporal patterns of a day (fragmentation, transitions, bouts, sleep timing) form distinct behavioral types beyond time budgets?
- Are demographic groups behaviorally homogeneous, or diverse within themselves?
- Does free-living behavior form discrete types, or does it vary continuously across clustering methods (GMM, hierarchical, HDBSCAN)?
- Prior clustering studies rarely test demographic association formally and none measure within-group diversity.

3 Objectives & Questions

Objective: *To identify interpretable 24-hour behavioral phenotypes and examine whether age and gender are associated with them.*

RQ1 **Do distinct phenotypes emerge from unsupervised analysis of 24-hour profiles?**

RQ2 **Is phenotype membership associated with age, gender, or their interaction?**

RQ3 **Does behavioral diversity within subgroups differ across age and gender?**

Contributions

- Formal association tests: χ^2 with Cramér's V
- Within-group diversity via Shannon entropy
- Time allocation vs. phenotype-level differences

4 Dataset

Age group	Female	Male	Total
18 to 29	26	15	41
30 to 37	25	12	37
38 to 52	30	7	37
53+	15	17	32
Total	**96**	**51**	**147**

CAPTURE-24: A free-living wrist-accelerometer dataset containing **24-hour** activity recordings from adults aged **18–79**, with activities labeled using wearable-camera images and time-use diaries.

5 Methodology

Daily activity and sleep profiles from free-living wrist-worn accelerometers were summarized into 19 features and clustered into three behavioral phenotypes, then tested against age and gender, with Shannon entropy measuring behavioral diversity within each demographic group.

1 . Data
CAPTURE-24: 151 participants, 147 kept (≥75% of the day annotated)

→ **2. 24-h day profile**
Recordings folded onto a 1,440-minute day; gaps imputed (30.3% → 25.0% unknown)

→ **3. 19 features**
Sleep (6) · Activity (8) · Temporal (5); circular sin/cos clock times

→ **4. Clustering**
GMM, Ward hierarchical and HDBSCAN; k = 2–8 by silhouette

→ **5. Statistics**
χ^2 + Cramér's V (age, gender);
Shannon entropy $H = -\Sigma\, p_j \log_2 p_j$

6 Results

A · THREE PHENOTYPES; DIVERSITY RISES WITH AGE

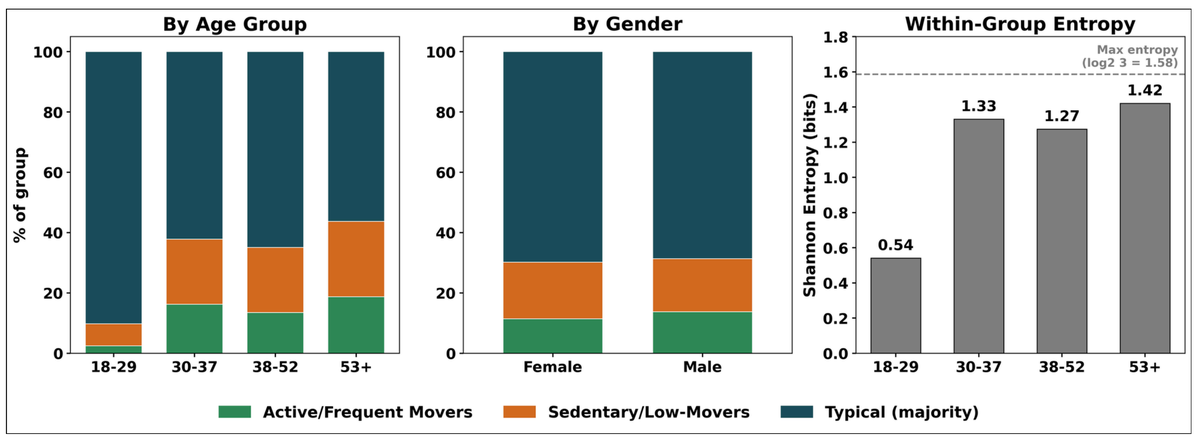


Fig:1 Phenotype mix by age and gender (GMM, k = 3) and within-group Shannon entropy (max = 1.585 bits).

Three phenotypes (Active Movers, Sedentary-Movers and Typical) were found. Typical dominates every group, making up **90%** of ages **18-29** but only **56** to **65%** of older groups. Men and women have nearly identical mixes. Within-group entropy rises from **0.54** bits **(18-29)** to **1.42** bits **(53+)**, close to the **1.585**-bit maximum, so older adults are the most behaviorally diverse.

B · GENDER GAPS LIVE IN TIME BUDGETS

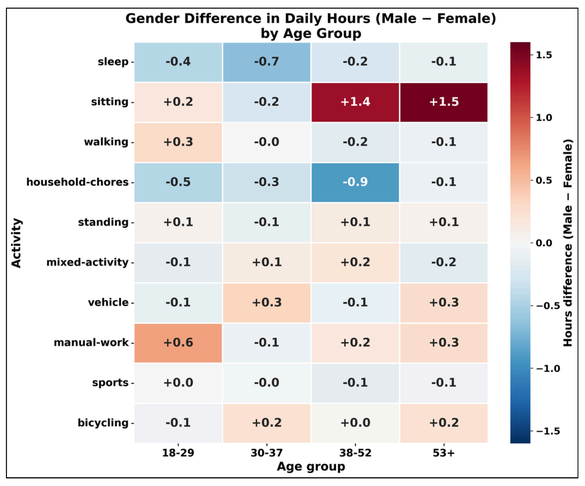


Fig:2 Male – female mean daily hours by age group.

- **Sitting:** +1.5 h more per day for men aged 53+
- **Chores:** −0.9 h less per day for men aged 38–52
- **Phenotypes:** about 70% Typical, 18% Sedentary, 12% Active, nearly identical for both genders

C · DAILY PROFILES: SLEEP BLOCKS SCATTER WITH AGE

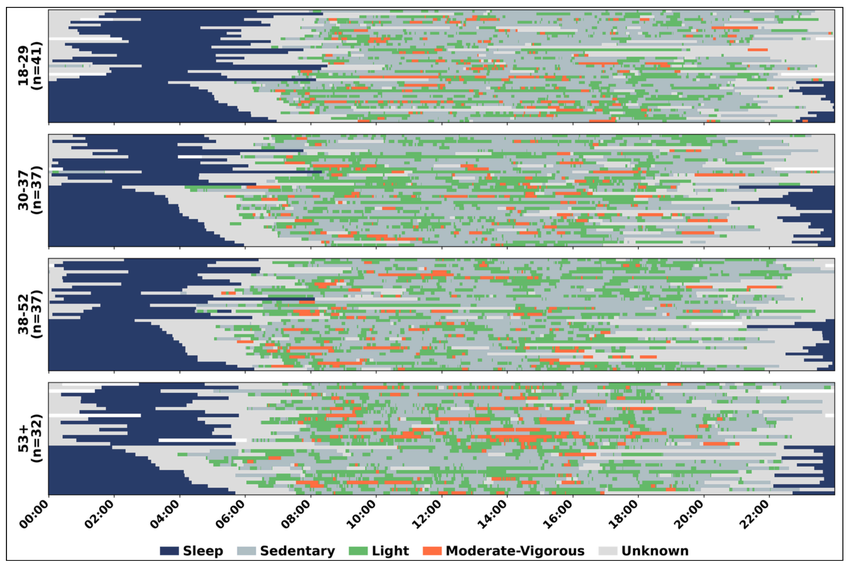


Fig:3 One row per participant. Overnight unknowns reflect camera removal.

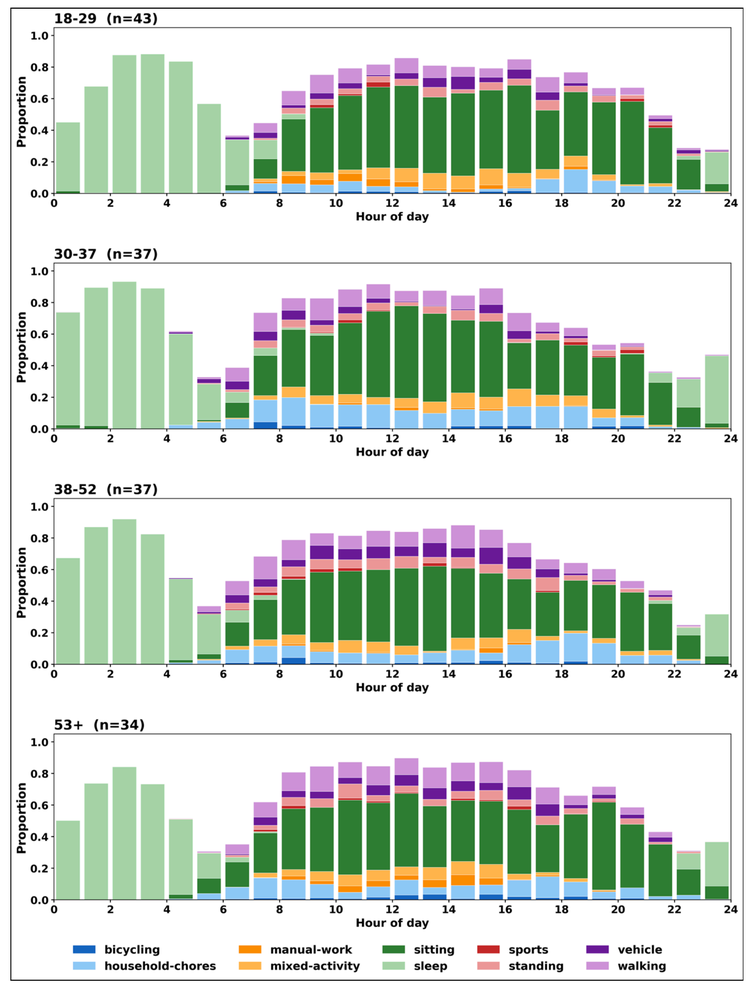


Fig:4 Mean activity hourly proportion by age group

1. All age groups show sleep-dominant nights, a 6–8 AM wake-up transition, and sitting-dominant days.
2. Waking starts earlier with age: about 6 AM for 53+ versus 8 AM for ages 18–29.
3. Older adults show more midday walking and chores; younger adults show more late-day sport.

7 Discussion

- Gender shapes what fills the day; age shapes how the day is organized.
- Adults **53+** spread across all phenotypes, so age alone says little about routine.
- Single **~24 h** recordings and **~25%** unknown minutes limit sleep estimates.
- Behavioral interventions for older adults may need to be individualized rather than group-targeted.
- Young adults (18-29) are the most uniform group, with 90% in one phenotype (entropy **0.54** bits), likely reflecting shared structure such as university schedules.

8 Conclusion & Future Work

Age moderately structures behavior, while gender does not. Gender differences appear in how time is allocated, not in the structure of the day. Interventions for older adults may need to be individualized rather than group-targeted.

Future Work:

Scale up: apply the pipeline to large cohorts (UK Biobank, NHANES, China Kadoorie Biobank) for narrower age bands, balanced gender groups and entropy confidence intervals.

- Multi-day recordings: separate stable behavioral traits from single-day variation, including weekdays vs. weekends.
- Richer features: use MET-weighted intensity and dynamic time warping on the raw 1,440-minute profile to split the large Typical group.
- Validation: test whether the phenotypes replicate on the held-out CAPTURE-24 split, and use a random forest with SHAP to show which features separate age groups.

References

[1] Doherty et al., "Large scale population assessment of physical activity using wrist worn accelerometers: the UK Biobank study," PLoS ONE, 2017. [2] Willetts et al., "Statistical machine learning of sleep and physical activity phenotypes from sensor data" Sci. Rep., 2018. [3] Chan et al., "CAPTURE-24: A large dataset of wrist-worn activity tracker data collected in the wild," Sci. Data, 2024. [4] Shim et al., "Wearable-based accelerometer activity profile as digital biomarker of inflammation, biological age, and mortality using hierarchical clustering analysis in NHANES" Sci. Rep., 2023.

Full paper under review at ABC Conference 2027

Contact: mdsahriar182@gmail.com